Smart Queue Management System

A **Smart Queue Management System (SQMS)** is a web or mobile application that helps businesses like banks, hospitals, government offices, and service centers **manage customer queues efficiently**. The system reduces wait times, improves customer experience, and optimizes service flow.

Key Features & Functionalities:

1. User Roles & Authentication

- **Admin:** Manage services, view analytics, and configure queue rules.
- **Staff:** View and call customers from the queue.
- **Customers:** Join queues remotely, track queue progress, and get notifications.

2. Virtual Queue & Ticketing

- Customers can take a **digital ticket** via the mobile app or web.
- Walk-in customers can scan a **QR code** at the service center to join the queue.
- Estimated wait time is displayed in real time.

3. Queue Status & Notifications

- Customers receive live updates on their position in the queue.
- SMS or push notifications alert them when their turn is near.
- Staff can update queue status (e.g., serving, completed, canceled).

4. Multi-Branch Support

- Customers can select a nearby branch before joining a queue.
- Admins can manage multiple locations from a central dashboard.

5. Appointment Scheduling (Optional)

- Users can book appointments in advance to avoid waiting.
- Integration with Google Calendar or Outlook for scheduling.

6. Analytics & Reports

- Average wait time per service.
- Peak hours and busiest days analytics.
- Performance tracking for service staff.

Technologies to Use:

- **Frontend:** React (for web) / React Native (for mobile).

- **Backend:** Django or Node.js with Express.
- **Database:** Firebase, PostgreSQL, or MongoDB.
- **Notifications:** Twilio (SMS), Firebase Push Notifications.
- **Authentication:** Firebase Auth, OAuth, or JWT.
- **Hosting:** AWS, Vercel, or Firebase Hosting.

Project Deliverables:

1. **Software Requirement Specification (SRS)**
2. **Use Case Diagrams & UML Models**
3. **Database Design (ERD, Schema)**
4. **Wireframes & UI/UX Design**
5. **System Architecture & API Design**
6. **Development & Implementation**
7. **Testing & Deployment Plan**
8. **Final Documentation & Presentation**